

一、選擇題 (單選) 30 題，每題 2 分，共 60 分，請在每題的選項內選擇最適當的答案。

注意：答錯倒扣 0.5 分，扣至零分為止。(不答不倒扣)

※ 注意：請用 2B 鉛筆作答於答案卡，並先詳閱答案卡上之「畫記說明」。

1. ☐ Which of the following data structures provides $O(1)$ average-time complexity for insertion, deletion, and search? (A) Binary search tree (B) Balanced binary search tree (C) Hash table (D) Sorted array (E) Heap.
2. ☐ Which of the following best describes the worst-case time complexity of QuickSort? (A) $O(n)$ (B) $O(n \log n)$ (C) $O(\log n)$ (D) $O(n^2)$ (E) $O(n^3)$.
3. ☐ What is the time complexity of inserting an element into a binary heap? (A) $O(1)$ (B) $O(n)$ (C) $O(\log n)$ (D) $O(n \log n)$ (E) $O(n^2)$
4. ☐ Which data structure is most appropriate for checking if an expression has balanced parentheses (e.g., $((a+b)*c))$? (A) Queue (B) Stack (C) Binary Search Tree (D) Hash Table (E) Graph.
5. ☐ Which of the following sorting algorithms is not stable? (Meaning relative order of equal elements is not guaranteed to be preserved). (A) Insertion Sort (B) Merge Sort (C) Bubble Sort (D) QuickSort (E) none of the above.
6. ☐ What is the prerequisite for performing a Topological Sort on a graph? (A) The graph must be undirected. (B) The graph must be a tree. (C) The graph must be a Directed Acyclic Graph (DAG). (D) The graph must be fully connected. (E) The graph must have non-negative edge weights.
7. ☐ Which of the following best describes the role of a cache memory in a computer system? (A) To permanently store program instructions (B) To reduce average memory access time (C) To manage virtual memory (D) To schedule CPU processes (E) To translate virtual addresses to physical addresses.
8. ☐ Which of the following CPU scheduling algorithms may cause starvation? (A) Round Robin (B) First-Come, First-Served (C) Multilevel feedback queue (with aging) (D) Priority scheduling without aging (E) Shortest remaining time first.
9. ☐ Which of the following best describes a context switch? (A) Switching between CPU cores (B) Changing page tables (C) Reloading cache blocks (D) Switching from kernel mode to user mode (E) Saving and restoring process execution state
10. ☐ In a pipelined processor, which situation most directly leads to a pipeline stall? (A) Instruction prefetching (B) Data hazard due to operand dependency (C) Branch prediction hit (D) Cache hit (E) Register renaming
11. ☐ Which CPU scheduling algorithm has the smallest average waiting time for a given set of processes? (A) First-Come, First-Served (FCFS) (B) Shortest Job First (SJF) (C) Round Robin (RR) (D) Multilevel Queue (E) Priority Scheduling.
12. ☐ What is the phenomenon called "Thrashing" in a virtual memory system? (A) The system spends more time paging (swapping pages in and out) than executing processes. (B) The hard disk fails due to excessive read/write operations. (C) A process enters an infinite loop. (D) Multiple processes are deadlocked waiting for resources. (E) The CPU pipeline is constantly flushed due to branch mispredictions.
13. ☐ What is the primary function of the Time to Live (TTL) field in an IPv4 header? (A) To determine the timestamp of when the packet was created. (B) To prioritize the packet for Quality of Service (QoS). (C) To ensure the packet is reassembled in the correct order. (D) To prevent packets from circulating indefinitely in a routing loop. (E) none of the above.
14. ☐ Which of the following is a characteristic of UDP (User Datagram Protocol) compared to TCP? (A) It guarantees

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- delivery of data. (B) It provides flow control and congestion control mechanisms. (C) It is connectionless and has a smaller header overhead. (D) It sequences packets to ensure they arrive in order. (E) It performs a three-way handshake before sending data.
15. ☐ In TCP congestion control, what is the main purpose of the slow start phase? (A) To avoid retransmissions entirely (B) To probe available network capacity exponentially (C) To maintain a constant congestion window (D) To prioritize latency-sensitive flows (E) To prevent receiver buffer overflow.
16. ☐ Which index structure is most used in relational databases to support range queries efficiently? (A) Hash index (B) Bitmap index (C) B+ tree (D) Inverted index (E) Heap file.
17. ☐ Which of the following concurrency control techniques may result in deadlocks? (A) Timestamp ordering (B) Two-phase locking (2PL) (C) Optimistic concurrency control (D) Multiversion concurrency control (MVCC) (E) Snapshot isolation.
18. ☐ Which of the following best explains why hash-based indexing is generally unsuitable for range queries? (A) Hash functions destroy ordering information (B) Hash indexes require more storage (C) Hash collisions slow down inserts (D) Hash indexes cannot support equality queries (E) Hashing increases I/O operations.
19. ☐ In write-ahead logging (WAL), which rule must be satisfied to ensure database recovery correctness? (A) Data pages must be written before log records (B) Log records are optional for read-only transactions (C) Checkpoints eliminate the need for redo (D) Log records must be written to stable storage before data pages (E) Logs are only needed for undo operations.
20. ☐ Which statement best captures a fundamental trade-off in database system design? (A) Strong consistency always improves performance (B) Indexing eliminates the need for query optimization (C) Higher isolation levels can reduce concurrency (D) Normalization always improves query speed (E) Replication removes the need for recovery mechanisms.
21. ☐ In the context of Distributed Database Systems, the CAP Theorem states that a distributed system can only simultaneously provide two out of which three guarantees? (A) Concurrency, Availability, Performance (B) Consistency, Atomicity, Partition Tolerance (C) Consistency, Authenticity, Persistence (D) Consistency, Availability, Partition Tolerance (E) Concurrency, Accuracy, Performance.
22. ☐ A relation R is in Boyce-Codd Normal Form (BCNF) if and only if: (A) For every functional dependency $X \rightarrow Y$, X is a superkey. (B) It is in 3NF and has no multivalued dependencies. (C) It is in 3NF and preserves all functional dependencies. (D) It contains atomic values only. (E) There are no partial dependencies.
23. ☐ Which of the following is a key advantage of containerization (e.g., Docker) compared to virtual machines? (A) Lower overhead due to sharing the host OS kernel (B) Stronger hardware-level isolation (C) No dependency on the host operating system (D) Elimination of resource contention (E) Guaranteed real-time performance.
24. ☐ Which of the following best describes the role of edge computing? (A) Centralizing all computation in large data centers (B) Replacing cloud computing entirely (C) Increasing storage capacity (D) Processing data closer to data sources to reduce latency (E) Encrypting data before transmission.
25. ☐ Which technology enables immutable and verifiable transaction records without relying on a central authority? (A) Distributed databases (B) Blockchain (C) Content delivery networks (D) Peer-to-peer file sharing (E) Federated learning.
26. ☐ Why are NoSQL databases often preferred in large-scale distributed applications? (A) They provide flexible schemas and horizontal scalability (B) They strictly enforce ACID properties (C) They eliminate network partitions (D) They guarantee strong consistency (E) They require less storage than relational databases.
27. ☐ In the context of Large Language Models (LLMs) like GPT, what is the primary function of the "Self-Attention"

mechanism in the Transformer architecture? (A) To compress the input data into a smaller vector. (B) To allow the model to weigh the importance of different words in a sequence relative to each other, regardless of their distance. (C) To generate random variations of the output to increase creativity. (D) To convert text into speech. (E) To store long-term memory of previous conversations on a hard drive.

28. ☐ What is the core principle of the “Zero Trust” security architecture? (A) “Never trust, always verify”—no user or device is trusted by default, even if inside the corporate network. (B) All internal traffic is trusted, but external traffic is blocked. (C) Using only biometric authentication for all access. (D) Isolating the network from the internet completely. (E) Relying solely on firewalls to protect the network perimeter.
29. ☐ What is the defining characteristic of “Serverless Computing” (e.g., AWS Lambda)? (A) There are physically no servers involved in the process. (B) The developer must manually patch and manage the operating system of the server. (C) The cloud provider dynamically manages the allocation of machine resources, and pricing is based on the actual amount of resources consumed by an application execution, not pre-purchased capacity. (D) It is a peer-to-peer network where every client is also a server. (E) Use commodity-grade PCs to replace high-cost servers.
30. ☐ In the field of Generative AI, what does “RAG” (Retrieval-Augmented Generation) refer to? (A) A method to generate random answers to prevent overfitting. (B) A technique to reduce the size of the model by removing unused neurons. (C) A training process that relies solely on reinforcement learning from human feedback. (D) A technique that optimizes the output of an LLM by referencing an authoritative knowledge base outside its training data before generating a response. (E) An AI agent learns to make decisions by interacting with an environment, taking actions, and receiving rewards or penalties.

二、問答題，共 40 分。

※ 本大題請於試卷內之「非選擇題作答區」標明題號依序作答。

1. The following shows an array-based implementation of ADT Stack that stores **non-duplicated** integers. NOTE: The Stack class has **NO member function of data swapping!!** All item switching or swapping operations must be implemented by yourself!!

```
const int MAXSIZE = 20; // the maximal stack size

class Stack
{
public:

    // insert an integer into the stack
    int push(int data);

    // reverse the order of the integers in the stack
    void reverse();

    // rank the integers in the stack in ascending order
    void rank();

    ...
private:
    int items[MAXSIZE]; // the array of the stacked integers
    int top; // the index of the top item, the initial is -1
};
```

- (a) (10 points) Write the `Stack::push()` function that inserts an integer into the stack. Note that the function returns -1 if either the stack is full or the input integer has already existed in the stack; otherwise, it returns 0.

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- (b) (10 points) Write the `Stack::reverse()` function that reverses the sequence of the stacked integers.
- (c) (20 points) Write the `Stack::rank()` function that ranks the stacked integers in ascending order. In other words, the top is the smallest integer. You **MUST** employ the bubble sort algorithm to rank the data!!

試題隨卷繳回